# Mathematics Method for Physics–Lecture x

Preliminaries of complex numbers

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Start up saying

## 1 Linear spaces

The objects in functional analysis often form a set that has a structure called the linear space, or vector space.

Let $\mathbb{K}$ be the scalar field, which throughout our course will be $\mathbb{R}$ or $\mathbb{C}$.

**Definition 1.** A linear space $X$ over $\mathbb{K}$ is a set together with an operation $+ : X \times X \to X$ called "addition" and scalar multiplication $\alpha u \in X$, $\forall \alpha \in \mathbb{K}$, $u \in X$.

## 2 Complex numbers

**Definition 2.** The set of complex numbers, denoted by $\mathbb{C}$, is defined as

$$\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}, i^2 = -1\}.$$